

ASHUTOSH KUMAR

Associate Software Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Associate Software Engineer with 1+ year of experience building secure, scalable full-stack apps using Node.js, NestJS, React.js, TypeScript, and PostgreSQL. Specialized in API Gateway design, Keycloak SSO, and JWT/DPoP security, with real-time systems using Kafka and WebSockets. Skilled in RBAC, PKI licensing, and Kubernetes deployments.

CORE SKILLS

Languages & Frontend: JavaScript, TypeScript, React.js, Redux Toolkit, HTML5, CSS3, Tailwind CSS

Backend & APIs: Node.js, NestJS, Express.js, REST APIs, Microservices Architecture, API Gateway

Auth & Security: OAuth2/OIDC, JWT (RS256/Ed25519), DPoP, Keycloak SSO, RBAC, PKI Licensing, M2M Auth

Real-Time & Streaming: Apache Kafka, WebSockets, Socket.IO, RTSP, HLS, FFmpeg, ROI Configuration

Databases: PostgreSQL, MongoDB, Redis, TypeORM, SQL, Schema Design

DevOps & Tools: Docker, Kubernetes, Git, GitHub, Postman, Swagger, CI/CD Basics

PROFESSIONAL EXPERIENCE

Evolve Cortex — Associate Software Engineer (Full-Stack Developer)

July 2025 – Present | Onsite, Mumbai (Trainee till Jun 2026; promoted to Associate Software Engineer Jul 2026)

- Designed a dual-deployment architecture (standalone + multi-tenant SaaS) with a shared, global Auth microservice (Keycloak) and API Gateway + Licensing backend reused across three products — HEX, DFX, WebX — under CAMS-EYE360.
- Designed an enterprise API Gateway with a multi-layer security pipeline (License Guard → RS256 JWT Guard → CSRF Guard → Proxy), isolating 6+ backend microservices from direct external access.
- Architected a cookie-first BFF authentication system with httpOnly JWTs, DPoP proof-of-possession, and Redis-backed refresh-token rotation — rated 8.5/10 in an internal security audit.
- Built a secure cross-product SSO handshake protocol and integrated Keycloak SSO with dual auth flows (Authorization Code for web, Direct Password Grant for the Flutter mobile app).
- Engineered a per-tenant licensing system using Ed25519 PKI, AES-256-GCM encrypted credential storage, and TOTP-based device binding.

KEY PROJECTS

CAMS – EYE 360 (AI-Vision Camera & Alert Management System)

React.js, NestJS, TypeScript, PostgreSQL, Redis, MongoDB, Apache Kafka, WebSockets, RTSP/HLS, FFmpeg, Docker, Kubernetes, Keycloak, Ed25519 PKI

- Architected a microservice platform (API Gateway, Auth BFF, Site Configuration, Project/Site Creation, SOC, Ticketing, Vendor Portal) behind a single-entry gateway with License Guard → JWT/DPoP Guard → CSRF Guard → Proxy.
- Built a per-tenant licensing system using Ed25519-signed licenses, environment-aware machine fingerprinting (K8s/Docker/local), heartbeat anti-rollback checks, and fail-closed startup validation.
- Designed a zero-trust BFF auth layer with Keycloak PKCE, httpOnly JWTs, DPoP-bound tokens, Redis-backed refresh rotation, JTI revocation, and reuse/replay detection.
- Implemented cross-product SSO (WebX → HEX) via atomic Redis single-use handoff tokens and Keycloak token exchange across independent realms.
- Engineered a real-time SOC pipeline (Kafka ingestion, Socket.IO fan-out, alert lifecycle) and camera/AI ops flows — RTSP → FFmpeg → HLS streaming, ROI configuration, and rights-gated React modules.

Resource Management Portal (RMG) – Lead & Recruitment Workflow System

NestJS, React.js, PostgreSQL, TypeScript, Redux Toolkit

- Built a full-stack recruitment workflow system (lead → requirement → resource) with JWT auth, RBAC, and an analytics dashboard.

EDUCATION

B.Tech, Computer Science and Engineering — GL Bajaj Institute of Technology and Management, Greater Noida, UP | 2020 – 2024

Intermediate — Tomar Children School, Varanasi, UP | 2018 – 2019

Matriculation — Sant Atulanand Convent School, Varanasi, UP | 2017 – 2018